Homework 1: MDPs and LQRs

CS/Stat 184(0): Introduction to Reinforcement Learning

(Due September 29 at 11:59 pm ET)

0 Instructions

For each question in this HW, please list all your collaborators and reference materials (beyond those specified on the website) that were used for this homework. Please add your remarks in a "Question 0".

1 Gardening as MDP [50 points]

Suppose that you are a gardener with an occasional aphid problem. When aphid-free, your garden produces at a rate of 1 unit of value per day. However, when you have an infestation, it produces nothing, and it remains infested until you clean it up.

The probability that an infestation occurs depends on how many days since you worked in the garden: the chance of an infestation occurring t days later is $\frac{t}{3}$. For example, suppose you work the field on Sunday. If you don't work on Monday, the probability of the garden becoming infested is 0. If you don't work on Tuesday, the probability of the garden becoming infested is 1/3. If you don't work on Wednesday, the probability of the garden becoming infested (if it isn't already) is 2/3. If you don't work on Thursday, if your garden is not infested already, it will become infested.

Working in the garden costs α units when there is no infestation, and 2α units once there is one.

We will model this problem as an infinite horizon Markov Decision Process with discounting factor $\gamma \in (0, 1)$. Actions $\mathcal{A} = \{1, 0\}$ represent choosing to work in the garden or not. Either the garden is infested with bugs (B) or its state is given by the days since you've worked it in, up to three (since then it is guaranteed to become infested). Therefore, $\mathcal{S} = \{B, 0, 1, 2, 3\}$.

1. [8 points] Write down the transition model $P(s'|s, a)$ and reward function $r(s, a)$ of this MDP.
2. [10 points] For some $C \in \{0, \dots 3\}$, consider the threshold policy

$$\pi_C(s) = \begin{cases} 1 & s \geq C \text{ or } s = B \\ 0 & \text{otherwise} \end{cases}$$

   Derive an equation for $V^{\pi_C}(B)$ and $V^{\pi_C}(s)$ for $s \geq C$ in terms of $V^{\pi_C}(0)$. Then, derive an iterative equation for $V^{\pi_C}(s)$ for $s < C$ in terms of $V^{\pi_C}(0)$ and $V^{\pi_C}(s+1)$. Hint: Use the Bellman Equation.
3. [16 points] For $\gamma = \frac{2}{3}$ and $\alpha = 0.5$, find the C_* such that $V^{\pi_{C_*}}(s) \geq V^{\pi_C}(s)$ for all s and C.
4. [16 points] Show that $\pi_{C_\star}$ is the optimal policy. Hint: use Bellman Optimality.

2 The Bellman Equations for $Q_h^\star$ [10 points]

Now let us show that $Q_h^\star$ satisfies:

$$Q_h^\star(s,a) = r(s,a) + \mathbb{E}_{s' \sim P(\cdot|s,a)}\left[\max_{a'} Q_{h+1}^\star(s',a')\right].$$

for all h, s, a. The above equations are the Bellman equations for $Q_h^\star$.

In class, we argued (without proof) that there exists a "simultaneously" optimal policy, i.e. there exists a policy $\pi^\star$ such that $V_h^{\pi^\star}(s) = \max_\pi V_h^\pi(s)$ and $Q_h^{\pi^\star}(s,a) = \max_\pi Q_h^\pi(s,a)$, for all s, h. This allows us to use the notation $V_h^\star = V_h^{\pi^\star}$ and $Q_h^\star = Q_h^{\pi^\star}$ to accurately refer to the values of the simultaneously optimal policies.

1. [5 points] Prove that $Q_h^{\pi^\star}(s, \pi_h^\star(s)) = \max_a Q_h^{\pi^\star}(s,a)$.
   Hint: Recall that, by definition, $Q_h^{\pi^\star}(s, \pi_h^\star(s)) = V_h^{\pi^\star}(s)$. Also, you may find both the Bellman equations and the Bellman consistency equations to be helpful.
2. [5 points] Now complete the proof. Hint: You may find your answer to part 1 to be helpful.

3 Approximating an infinite horizon, discounted MDP with a finite horizon one [30 points]

We defined a finite horizon MDP by the tuple $\mathcal{M}_H = \{\mu, S, A, P, r, H\}$. Here, $r : S \times A \to [0, 1]$ is the reward function, where $r(s, a)$ is the instantaneous reward at any given timestep. Now consider the modification to this MDP to $\mathcal{M}_H = \{\mu, S, A, P, \overline{r}, H\}$, where we let the reward function depend on the current timestep. More precisely, we have that $\overline{r} : S \times A \times \{0, \dots H\} \to [0, 1]$, so that $\overline{r}(s, a, h)$ is the instantaneous reward at timestep h.

We let:

$$V_h^\pi(s) = \mathbb{E}\left[\sum_{t=h}^{H-1} \overline{r}(s_t, a_t, t) \middle| s_h = s\right],$$

where we have modified $V_h^\pi(s)$ to use our new reward function.

1. [5 points] Write down a dynamic programming algorithm which computes both $V_h^*(s)$ and $\pi_h^*(s)$. (You do not need to prove that your program is correct though you should be able see how to generalize the original algorithm.)

Let us now consider a discounted MDP, $\mathcal{M}_\gamma = \{\mu, S, A, P, r, \gamma\}$. Here r is the usual reward function, where $r(s, a)$ is the instantaneous reward at state-action pair (s, a). You will now specify a finite horizon MDP which approximates this infinite horizon discounted MDP in a precise sense.

2. [15 points] (Effective Horizon) We will now provide an integer H_ε such that for any sequence of numbers $x_0, x_1, \dots$, where each $x_t \in [0, 1]$, we have that:

$$\left| \sum_{t=0}^{\infty} \gamma^t x_t - \sum_{t=0}^{H_\varepsilon - 1} \gamma^t x_t \right| \le \varepsilon.$$

   In particular, show that it suffices to take:

$$H_\varepsilon = \left\lceil \frac{\ln\left[1/((1-\gamma)\varepsilon)\right]}{1-\gamma} \right\rceil$$

   Here, it may be helpful to use that $-\ln(x) \ge 1 - x$ for $x \ge 0$ (this inequality itself follows from $1 + x \le \exp(x)$).

Now you will define a finite horizon MDP $\mathcal{M}_{\overline{H}} = \{\overline{\mu}, S, A, \overline{P}, \overline{r}, \overline{H}\}$ which can be thought of as approximating $\mathcal{M}_\gamma = \{\mu, S, A, P, r, \gamma\}$. We have already specified that both MDPs share the same state and action spaces. Let $V_0^\pi(s)$ refer to the finite horizon, value function corresponding to the MDP $\mathcal{M}_{\overline{H}}$, and let $V_\gamma^\pi(s)$ be the discounted, infinite-horizon value function corresponding to the MDP $\mathcal{M}_\gamma$. It remains for you to specify $\overline{\mu}$, $\overline{P}$, $\overline{r}$ and $\overline{H}$. What we desire is to specify these so that for all s, and for all stationary policies π, we have that:

$$\left|V_\gamma^\pi(s) - V_0^\pi(s)\right| \le \varepsilon.$$

(recall that a stationary policy π does not choose actions dependent on the current timestep in the episode).

3. [6 points] Precisely, specify choices of $\overline{\mu}$, $\overline{P}$, $\overline{r}$ and $\overline{H}$, which are written in terms of only ε and any other relevant quantities from the MDP $\mathcal{M}_\gamma$ specification. Also, consider your answer to the previous question. (Hint: you are permitted to allow $\overline{r}$ to be non-stationary.).
4. [4 points] Give a brief argument as to why your choice implies the condition above; specifically, explain what motivated your choice $\overline{\mu}$ and $\overline{P}$.

It is natural to refer to $\overline{H}$ as the ε-effective horizon. The interpretation is that we have a found a finite horizon MDP which approximates the infinite horizon discounted MDP. In particular, dynamic programming in the finite horizon MDP you constructed leads to an ε-optimal value function in the infinite horizon discounted MDP. It is is interesting to note that this approach is nearly identical to the value iteration algorithm; the finite horizon dynamic program leads to a sequence of values nearly identical to the value iteration algorithm, differing only by a scale factor at each iteration (do you see why?).